

Training Deep Neural Networks to Meet Hard Prediction Quotas

Anonymous submission

Abstract

We study *transductive count constraints*: hard caps on how many test-set items a classifier may assign to a constrained class (a triage queue that admits only so many “urgent” cases), optionally with per-group local caps. Post-hoc clipping meets the cap but erodes accuracy by editing predictions, and dual-ascent training escalates an unbounded multiplier that can climb steeply and still stall short of feasibility. We propose **TraLO**, a bounded transductive loss whose penalty saturates and whose multiplier ratchets while the count is violated and freezes once it is met, with an optimizer reset and an undershoot hinge that restores constrained-class recall. Across three MedMNIST datasets and three backbones (MobileNetV3, RegNetY-400MF, ViT-B/16), TraLO matches the strong constraint-trained baselines across most of the grid and pulls ahead on constrained-class F1 wherever the count binds hard: on OctMNIST tight caps the lead is backbone-general and *grows with backbone capacity*, reaching +0.081 on the ViT, and it recurs, preliminarily, on the global-tighter asymmetric caps of DermMNIST. Against post-hoc clipping, the constraint-trained family including TraLO improves macro-F1 on every dataset and backbone by landing on the cap natively, the one exception being the dual baselines tying the clipper on OctMNIST with the ViT. We report the full regime map, wins and ties alike, with a component ablation.

1 Introduction

A surprising number of classification systems are deployed under a constraint not on which items get a label but on how many do. A clinical triage tool may escalate only a fixed fraction of scans to a specialist, and an inspection line can act on no more flagged items than it has inspectors. In each case the operational requirement is a *hard upper bound on the count* of predictions in a constrained class over the evaluation set. The items to be labeled are known in advance, so the constraint is naturally *transductive*: it is a property of the whole test-set prediction vector, not of any single example.

The pervasive way to honor such a cap is *post-hoc clipping*: train a normal model, rank the test items by their score for the constrained class, keep the top- K , and reassign the rest. This is trivial to implement and always meets the count, which is why it is the realistic deployment baseline. But it meets the count by editing predictions, often dozens of them, and every edit is a forced error on an item the model was previously confident about, so accuracy degrades. The other

end of the spectrum is *constrained training*: encode the cap as a constraint and solve the Lagrangian with dual ascent (Fioretto et al. 2020; Hounie, Ribeiro, and Chamon 2023; Chamon and Ribeiro 2020). In principle this is exactly the right formulation, but for a hard count the dual multiplier of a linear penalty climbs without bound. While the classifier is still learning, its gradient keeps re-inflating the constrained class, so the violation persists and the multiplier climbs every epoch. A one-epoch-warmup probe shows it climbing steeply while our bounded multiplier barely moves, for the same final count (Supp. Sec. A). The implementations remain competitive, but the constrained class is the part of the model most stressed by the escalation, and on some cells the dual never reaches feasibility at all (Sec. 5.4).

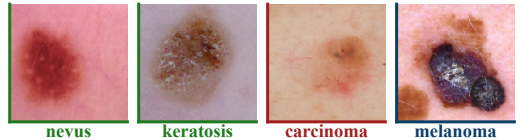


Figure 1: **DermMNIST lesions, benign to malignant** (frame color: green benign, red malignant, blue the *constrained* class). The task caps how many test images the model may label melanoma, the rare high-stakes class; TraLO meets that cap *during training* rather than clipping afterward. All three datasets we study have such a rare class.

We propose **TraLO** (Transductive Loss Optimization), a constrained-training method whose multiplier stays bounded by construction. Its transductive penalty is bounded (a rational-saturation term plus a bounded quadratic in the soft excess over the cap), the multiplier only ever moves upward while the count is violated and freezes the first time it is met, the Adam state is reset at first satisfaction to kill post-satisfaction descent momentum, and an undershoot hinge restores constrained-class recall once the cap is met. The result is a classifier that lands on the count by itself, with no prediction surgery.

Contributions.

- **A deployable formulation and method.** We cast the operational “ $\leq K$ predictions of class c ” requirement

as a transductive count constraint and give a bounded penalty with a freeze-at-satisfaction multiplier, optimizer reset, and undershoot hinge (Sec. 3).

- **A characterized quality win over the trained baselines.** Versus the much stronger constraint-trained baselines, TraLO matches constrained-class F1 across most of the grid and beats them wherever the count binds hard: on all three backbones, at OctMNIST tight caps, where the lead grows with backbone capacity and reaches +0.081 on the ViT transformer, and preliminarily, on two of the three backbones, at the global-tighter asymmetric caps of DermMNIST. We publish the full regime map including the large region where it is a statistical tie (Secs. 5.1–5.2).
- **A quantified accuracy tax for post-hoc clipping, and a family that avoids it.** Across all three datasets the constraint-trained family, TraLO included, improves macro-F1 over the field-default clipping baseline on every backbone including the ViT transformer, save a tie at OctMNIST×ViT (Sec. 5.3). TraLO delivers the gain with native satisfaction on every run in our grid and the fewest prediction edits at binding caps (Supp. Sec. C).
- **Mechanism and ablation.** We separate two roles: the bounded penalty makes the count reachable with a small, frozen multiplier, while the constrained-class quality gain is carried by the optimizer reset and undershoot hinge, isolated in a complete ablation (Supp. Sec. B).

2 Related Work

Lagrangian and dual constrained learning. Encoding requirements as constraints and solving with primal–dual methods is the classical recipe (Bertsekas 1982), brought to deep networks by Lagrangian-duality training (Fioretto et al. 2020) and given statistical footing by PAC constrained-learning theory (Chamon and Ribeiro 2020; Chamon et al. 2023). Resilient Constrained Learning (Hounie, Ribeiro, and Chamon 2023) adapts the constraint level to make hard requirements attainable, and PID-style dual updates stabilize the multiplier in safe RL (Stooke, Achiam, and Abbeel 2020), and Gallego-Posada et al. (2022) make the empirical case that constraints outperform fixed penalties on sparsity targets. Our critique of runaway escalation is therefore specific to the unbounded linear-penalty formulation: augmented-Lagrangian and resilient updates keep the multiplier better behaved, yet the resilient method we benchmark still winds up slowly on a hard count (Sec. 5.4). Closer to our output-level setting, constraints on a network’s predictions have been enforced during training by primal–dual updates (Nandwani et al. 2019) and by linear constraints on predicted label counts for weakly supervised segmentation (Pathak, Krähenbühl, and Darrell 2015), or guaranteed per prediction by construction through input-dependent parametrizations (Brosowsky et al. 2021) or an output-layer satisfiability solver (Goyal, Dumančić, and Blockeel 2024); Márquez-Neila, Salzmann, and Fua (2017) find hard enforcement feasible but no better than soft penalties, which motivates our bounded penalty. Constrained optimization has also been used to protect critical, under-represented classes (Sangalli et al. 2021), and group-wise constraints underpin fair classification (Zafar et al. 2017;

Agarwal et al. 2018), the trained analogues of our local caps. We use the two strongest dual-ascent methods, Fioretto-LDF (Fioretto et al. 2020) and Hounie-RCL (Hounie, Ribeiro, and Chamon 2023), as our constraint-trained baselines. Our point of departure is that a hard count cap is the setting where the dual multiplier escalates without bound; TraLO instead optimizes a bounded penalty and freezes the multiplier at satisfaction.

Rate-constrained learning. The closest classical relatives constrain prediction *rates*: coverage (the fraction of items predicted positive) as a dataset constraint (Goh et al. 2016), non-decomposable rate objectives through surrogate bounds (Eban et al. 2017), non-differentiable rate constraints through proxy-Lagrangian games (Cotter et al. 2019), and generalized rate metrics through multi-player schemes (Narasimhan, Cotter, and Gupta 2019). These bound *expected* rates over a distribution, inductively. Our requirement is a hard count on one specific test set: transductive, verified on the argmax predictions themselves, and paired with per-group local caps.

Post-hoc correction and resource allocation. Resource-constrained classification allocates a fixed budget over a trained classifier’s outputs at decision time, classically as a small assignment problem, most recently under exactly our local-plus-global constraint structure (Shifman et al. 2025). Other work in this line folds the budget into learning via adaptive cost-sensitive weights (Shifman et al. 2023) or predict-and-optimize ranking under stochastic capacity (Vanderschueren et al. 2024), steering expected costs rather than enforcing a hard count; learning-with-rejection (Cortes, DeSalvo, and Mohri 2016) prices abstention per instance, likewise without a count guarantee. In fair classification, post-processing likewise edits a trained model’s decisions with group-specific thresholds (Hardt, Price, and Srebro 2016). Decision-time allocation always meets the budget, but it is a deploy-time clipper: it edits the model’s predictions to do so. We use a greedy threshold heuristic and a linear-programming clipper (Shifman-LP, after the two-phase allocator of Shifman et al. 2025) as our post-hoc baselines, and show TraLO removes the accuracy tax they pay.

Cost-sensitive and imbalanced training. Reweighting or reshaping the loss changes which examples are emphasized: cost-sensitive thresholds (Elkan 2001), focal loss (Lin et al. 2017), class-balanced weights (Cui et al. 2019), label-distribution-aware margins (Cao et al. 2019), prior-corrected logits (Menon et al. 2021). None of it guarantees the prediction count; the realized number of constrained-class predictions remains a free outcome. TraLO targets the count directly.

Transductive and set-level constraints. Transductive inference reasons about a fixed unlabeled test set (Vapnik 1998; Joachims 1999), and soft set-level targets have long existed there: expectation regularization pushes predicted class proportions on unlabeled data toward a prior (Mann and McCallum 2007), and posterior regularization constrains posterior expectations (Ganchev et al. 2010). Our constraint is transductive in exactly this sense, a function of the entire test-set prediction vector, but it is *hard*: an explicit argmax count

budget on a single class, with per-group caps, enforced during training of a deep classifier.

3 Method

Problem. Let the (known) test set contain N items and let $p_\theta(y = c \mid x_i)$ be the model’s probability for the constrained class c on item i . A *global* count constraint requires the number of items predicted c to satisfy $\sum_{i=1}^N \mathbf{1}[\hat{y}_i = c] \leq K$. We parameterize the cap as a percentage L of the constrained class’s true test-set count, so L measures how much of the class’s actual demand the quota admits ($L30$: the model may assign the label to at most 30% as many items as truly carry it). *Local* caps also bound the count within each group g : $\sum_{i \in g} \mathbf{1}[\hat{y}_i = c] \leq K_g$ (the same percentage of the group’s own class count). Because the indicator is not differentiable, training uses the *soft count* $S = \sum_i p_\theta(y = c \mid x_i)$, while satisfaction is verified with the *hard count* from $\arg \max$.

A bounded transductive penalty. The defining choice in TraLO is that the constraint penalty is *bounded*. For a soft excess $E = \max(0, S - K)$ over a cap K , the per-constraint penalty is

$$\ell(E; K, \rho) = \underbrace{\frac{E}{E+K}}_{\text{rational saturation}} + \rho \underbrace{\frac{(E/K)^2}{1 + (E/K)^2}}_{\text{bounded quadratic}}, \quad (1)$$

where both terms saturate as $E \rightarrow \infty$ (the first to 1, the second to ρ). What actually drives training is the penalty’s *gradient* on the count; differentiating Eq. (1) through $E = \max(0, S - K)$ gives, for $S > K$,

$$\frac{\partial \ell}{\partial S} = \frac{K}{(E+K)^2} + \frac{2\rho}{K} \frac{E/K}{(1 + (E/K)^2)^2}. \quad (2)$$

Both the value and this gradient are uniformly bounded, and that bound is what makes the constraint pressure controllable.

Proposition 1 (Bounded, non-escalating pressure) *For $E \geq 0$ and $\rho \geq 0$, the penalty of Eq. (1) obeys $0 \leq \ell < 1 + \rho$, and its count-gradient lies in $[0, (1 + \rho)/K]$: strongest ($\rightarrow 1/K$) at the boundary $E=0$, and vanishing as $E \rightarrow \infty$. Hence with the multiplier frozen at λ , each cap adds at most $\lambda(1 + \rho)$ to the loss and at most $\lambda(1 + \rho)/K$ to the gradient norm, both independent of the violation size.*

(Proof: elementary calculus, Supp. Sec. A.) A linear penalty λE instead contributes a constant gradient and a quadratic one $\rho(E/K)^2$ a gradient $2\rho E/K^2$ unbounded in E , so both force the multiplier to escalate to keep pace with cross-entropy (Fig. 2). Proposition 1 bounds the constraint pressure and keeps it from escalating with the violation; that a *small* frozen multiplier then suffices to hold the count ($\lambda_g \leq 0.09$ across our grid) is an empirical finding, not part of the proposition, as are the constrained-class quality gains, which the ablation attributes to the optimizer reset and hinge (Supp. Sec. B) rather than the penalty shape. The total objective combines cross-entropy with the global and local penalties (and, after satisfaction, the undershoot hinge introduced below),

$$L_{\text{total}} = L_{\text{CE}} + \lambda_g \ell_g + \lambda_l \ell_l, \quad (3)$$

where $\ell_g = \ell(E; K, \rho)$ is the global-cap penalty and ℓ_l sums Eq. (1) over the group caps; every cap (the global one and each group cap) carries its own multiplier, so $\lambda_l \ell_l$ abbreviates $\sum_g \lambda_{l,g} \ell(E_g; K_g, \rho)$.

Multiplier ratchet and freeze. Rather than dual ascent, λ is *ratcheted*: it increases by a fixed step each epoch *only while the hard count violates the cap*, and is *frozen* the moment the count is first satisfied. The bounded penalty means a small frozen multiplier suffices (empirically $\lambda_g \leq 0.09$), so cross-entropy is never swamped. Multipliers start at 0.05. The quadratic weight ρ ramps linearly (from 5 toward 100) until first satisfaction and then also freezes. Because λ and ρ both stop moving at first satisfaction, TraLO is really a two-phase procedure: while the count is violated it *searches* for a sufficient multiplier by ratcheting λ (the bounded gradient of Eq. (2) keeps that search ending at a small λ), and once the count is met it *settles*. Formally, for every epoch $t \geq T_{\text{sat}}$ (first satisfaction) the coefficients $\lambda_g, \lambda_l, \rho$ are constant, so Phase 2 is plain SGD on one fixed objective $F(\theta) = L_{\text{CE}} + \lambda_g \ell_g + \lambda_l \ell_l$. Under the standard assumptions for smooth nonconvex SGD it therefore converges to a first-order stationary point of F (Ghadimi and Lan 2013; Bottou, Curtis, and Nocedal 2018); ℓ meets the gradient bound of Prop. 1 and is smooth except on the measure-zero boundary $\{S_c = K_c\}$, which a softplus in place of the ReLU would remove. A dual or augmented-Lagrangian update instead changes the objective every step, so its iterate chases a moving target and inherits no such guarantee without extra assumptions on the multiplier sequence. What Proposition 1 does *not* give is feasibility from the loss: since ℓ saturates, no finite λ makes it an exact penalty (Bertsekas 1982), so TraLO does not enforce the hard count through ℓ alone. It need not: the ratchet raises λ only until the $\arg \max$ count is met, and the post-hoc step (Alg. 1) closes any residual soft-to-hard gap. The guarantee is thus one of *stability*, a bounded and non-escalating penalty on a fixed Phase-2 objective, not penalty-driven feasibility.

Optimizer reset at satisfaction. At the first epoch the count is met, we reset the Adam optimizer state. This removes the accumulated descent momentum that would otherwise keep pushing the soft count far below the cap (oversatisfying at the cost of constrained-class recall). The reset lets the frozen-multiplier objective settle near the boundary instead of overshooting inward. We motivate the reset empirically; the ablation (Supp. Sec. B) isolates its effect, and we do not claim to derive it.

Undershoot hinge. A satisfied cap can still leave the model under-predicting the constrained class, which costs recall. Concretely, each cap c adds $\lambda_c \beta \max(0, K_c - S_c)/K_c$ to the loss with $\beta=0.5$. It is zero whenever the soft count sits at or above its cap, so it can never re-violate. The effect: after satisfaction this one-sided hinge gently pushes the soft count back up toward (but not over) the cap whenever it has undershot, restoring constrained-class recall without re-violating the count. Supp. Sec. B shows the hinge is one of the two components that carry the effect, together with the optimizer reset.

Algorithm 1 TraLO training loop

```
1: warm up  $\theta$  on  $L_{CE}$  alone; cache the warmup model
2: for each constraint epoch do
3:   update  $\theta$  on  $L_{CE} + \lambda_g \ell_g + \lambda_l \ell_l$  (plus the hinge once
      satisfied) (Eq. 3)
4:   recount hard predictions on the test inputs
5:   if a cap is violated then
6:     ratchet that cap's  $\lambda$  up one step; advance  $\rho$ 
7:   else if first time all caps are met then
8:     freeze  $\lambda$  and  $\rho$ ; reset the Adam state
9:   end if
10:  stop after 5 consecutive satisfied epochs
11: end for
12: post-hoc verification (flip borderline predictions) if
    needed
```

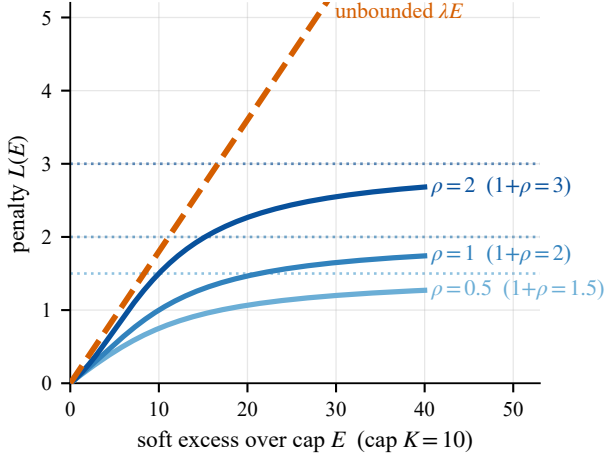


Figure 2: **A bounded penalty keeps the dual multiplier small.** The per-constraint penalty of Eq. (1) for several ρ (blue) flattens to its ceiling $1+\rho$ (dotted); a linear penalty λE (dashed) has none. Analytic illustration ($K=10$); curves labelled at the right edge.

Training schedule. Training has a warmup phase (cross-entropy only, to a stable initialization that is cached and reused) followed by the constraint phase under Eq. (3). A post-hoc verification step is available to enforce a hard count if a run has not landed natively; for TraLO it is essentially never needed (Supp. Sec. C). Algorithm 1 summarizes the loop.

4 Experimental Setup

Datasets. We use three MedMNIST v2 (Yang et al. 2023) classification tasks with their official splits, each with a naturally rare class that makes a count cap meaningful (Figure 1 shows DermMNIST examples): **TissueMNIST** (8 classes, constrained class GE, $\approx 7\%$ of the test set), **DermMNIST** (7 skin-lesion classes from HAM10000 (Tschandl, Rosendahl, and Kittler 2018), constrained class melanoma, with per-group local caps), and **OctMNIST** (4 retinal-OCT classes,

constrained class drusen). Caps are set as a percentage L of the constrained class’s true test-set count (Sec. 3); for the local-cap (asymmetric) study we also vary the group cap G independently of L . The percentage parameterization uses the class’s true count only to standardize how hard the cap binds across datasets; the method itself receives an exogenous integer K , exactly as a deployed capacity would be given.

Backbones. The headline grid uses three backbones, MobileNetV3 (Howard et al. 2019), RegNetY-400MF (Radosavovic et al. 2020), and a ViT-B/16 transformer (Dosovitskiy et al. 2021), to show the findings are not CNN-specific. All backbones use standard ImageNet-style training conventions (He et al. 2016).

Baselines. Two *post-hoc clippers*: a greedy threshold **Heuristic** and a linear-programming clipper **Shifman-LP** (Shifman et al. 2025). Two constraint-trained dual-ascent methods: **Fioretto-LDF** (Fioretto et al. 2020) and **Hounie-RCL** (Hounie, Ribeiro, and Chamon 2023). We also report **TraLO-bounded**, our own ablation with the undershoot hinge switched off, used both as a method and as the within-cell control in Supp. Sec. B.

Baseline adaptations. The two trained baselines were built for a different setting, so we re-instantiated each on the transductive count task and ran it inside TraLO’s exact harness: the same cached warmup, the same 300-epoch budget and stop rule (five consecutive feasible epochs), the same gradient clipping, and the same post-hoc verification step at the end. Two details keep the comparison fair. We point each method’s constraint gradient at the unlabeled test set, since that is where the count is enforced, whereas the original papers target a training or population expectation. And we select each method’s reported checkpoint on the constraint-excess axis (best feasible, else lowest excess), never on F1, so no baseline can quietly keep its best-F1 epoch. **Fioretto-LDF** is otherwise unchanged: a linear penalty $\lambda_c S_c$ with per-constraint subgradient ascent $\lambda_c \leftarrow \lambda_c + \eta \max(0, S_c - K_c)$ on the test counts. **Hounie-RCL** keeps its primal/slack/dual updates, with the constraint residual read as $(S_c - K_c)/N$. The one substantive fix in our port is to normalize its primal count-gradient by the sample count so the primal step and the dual update share a scale. Otherwise the primal term is $N \times$ too strong and over-suppresses the capped class. Its step sizes are fixed a priori, not tuned per cell, and swept only to confirm fairness (Supp. Sec. A). **Shifman-LP** is post-hoc only: a global-and-local linear program that allocates the capped labels on the warmup model’s test-set softmax, using an identity misclassification cost rather than the general cost matrix of the original two-phase allocator (Shifman et al. 2025).

Frozen recipe. All methods share one protocol, fixed before the final runs: ImageNet-pretrained backbones, 224×224 inputs, batch size 64, no augmentation, Adam with learning rate 10^{-4} during warmup and 5×10^{-6} in the constraint phase; warmup = 50 epochs (cross-entropy only), constraint phase = 300 epochs, seeds 1–4 (all reported, no seed selection). Per-method constraint steps were fixed

a priori, not tuned per cell: dual step 0.005 (Fioretto-LDF) and dual and relaxation steps 0.01 (Hounie-RCL) on TissueMNIST and DermMNIST, and 0.01 and 0.1 respectively on OctMNIST; ratchet step 0.002 with hinge weight $\beta=0.5$ (TraLO). A step-fairness sweep (Supp. Sec. A) confirms the OctMNIST result is insensitive to these choices. The headline grid is *paper_final*: 3 backbones $\times$ 3 datasets $\times$ 9 cap levels $\times$ 6 methods $\times$ 4 seeds = 1944 runs. Models are implemented in PyTorch with torchvision backbones and MedMNIST v2 loaders, and each run trains on a single NVIDIA GPU. Code, the configuration generators, and the full per-run results corpus accompany the submission as the Code and Data Supplement.

Metrics. Our *primary* metrics are quality: **constrained-class F1 (cc-F1)**, the quality of the capped class itself and the quantity the constraint most directly puts at risk, and **macro-F1** for overall quality. As *secondary* deployment diagnostics we also report **native satisfaction** (the fraction of runs whose final model meets the global count with no post-hoc edit) and **flips** (the number of test predictions the post-hoc verification step overwrites to enforce the global and per-group caps; 0 when the run needs no adjustment); we treat these as deployment *properties* rather than headline metrics, since every deployable method meets the hard count one way or another (Supp. Sec. C). Gaps versus baselines are *paired* by seed: because the seeds are matched, we summarize a method’s advantage by its seed-winrate and a paired Wilcoxon test rather than by marginal $\pm$ std bars, which would understate a paired effect when a baseline is seed-unstable but its per-seed gap to TraLO is consistent.

5 Results

5.1 Constrained-class quality vs. the trained baselines

We start with constrained-class F1 (cc-F1), the quality of predictions on the very class the cap restricts. Against the constraint-trained baselines the summary is a tie almost everywhere, and a win where the cap binds hard. Table 1 shows the symmetric grid on MobileNetV3: among the trained methods TraLO is best-or-tied in 7/9 cells, but at equal feasibility the differences are within seed noise (paired Wilcoxon, not significant): a genuine tie on tissue and derm. The tie also holds away from the grid, at a tighter ($L40$) and a looser ($L60$) cap on RegNet.

The hard regime: OctMNIST tight caps. The one place TraLO pulls clearly ahead of the trained baselines is OctMNIST at tight caps, where the cap binds hard but the drusen class is still discriminable. There, TraLO leads the best trained baseline on cc-F1 at every backbone, and the margin *grows with backbone capacity*: $+0.016/+0.022$ on MobileNetV3 ($L30/L40$), $+0.035/+0.021$ on RegNet, and $+0.081/+0.051$ on the ViT-B/16 transformer, at a 4/4 seed winrate on every backbone; pooled over the six tight-cap cells TraLO wins all 24 paired seeds (paired Wilcoxon $p < 10^{-6}$), and the same cells improve macro-F1 at five of the six (mean $+0.012$, up to $+0.028$ on the ViT; Table 2). Figure 3 plots the full sweep: on a representative backbone (top) the six

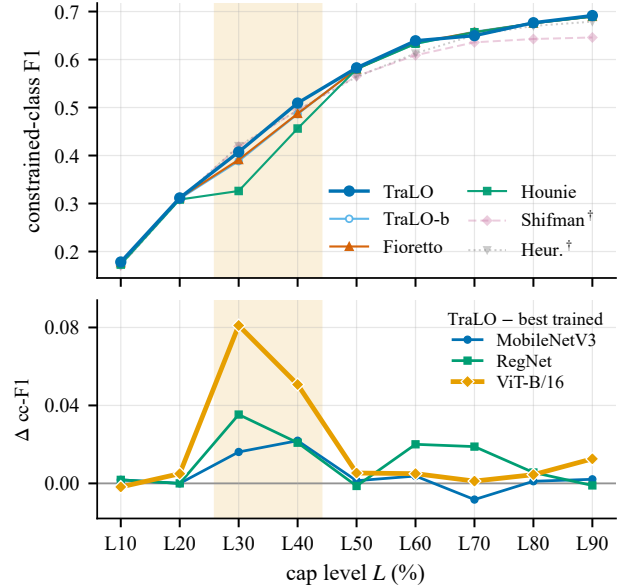


Figure 3: **The OctMNIST tight-cap win is backbone-general and grows with backbone capacity.** *Top*: constrained-class F1 of all six methods (MobileNetV3, mean over $n=4$ seeds); the curves nearly coincide, so the win is invisible at this scale (shaded band = tight-binding regime; † faded post-hoc clippers post higher raw cc-F1 by editing). *Bottom*: TraLO minus the best constraint-trained baseline, paired by seed, per backbone; the same inverted-U appears on all three, peaking at $+0.081$ for ViT-B/16.

methods nearly coincide, but the per-backbone gap (bottom) reveals the tight-binding win and its honest inverted-U: near-zero at $L10/L20$ and at loose caps, positive only where the cap binds. The post-hoc clippers’ higher raw cc-F1 in the band is bought by editing predictions (Supp. Sec. C).

Robustness of the OctMNIST win. Three checks. (i) *Backbone-general*: in leave-one-backbone-out cross-validation (define the winning band on one backbone, test on the held-out one), every held-out backbone confirms the win at 100% winrate (pooled held-out gap $+0.049$). (ii) *Re-derivable without the gap*: the winning band is recovered by a rule that never looks at the TraLO-vs-baseline gap, “binding (a clipper must flip ≥ 65 predictions) and discriminable (constrained-class F1 $\in [0.35, 0.55]$)”, under which HIT cells win 100% and MISS cells 0–40%. (iii) *Honest shape*: the effect is an inverted-U in tightness, not “tighter is always better,” and it is specific to OctMNIST; the same rule does not transfer to tissue/derm. We claim this as the one *backbone-general* cc-F1 win; the regime map (Sec. 5.2) adds one preliminary win at global-tighter asymmetric caps (on two of the three backbones), and a broad tie elsewhere, not a blanket cc-F1 advantage.

Table 1: **Constrained-class F1 on the symmetric grid** (MobileNetV3, mean over seeds 1–4; $\pm$ across-seed std). **Bold** = best, underline = second-best among the constraint-trained methods; † = post-hoc clippers, shown for reference. Among the trained methods the differences sit within seed noise (a paired-Wilcoxon tie); the marginal std here is shown for transparency, but the paired win uncertainty is the seed-winrate of Table 2.

Data	Cap	Post-hoc †		Constraint-trained			
		Heur.	Shifman	Fioretto	Hounie	TraLO-b	TraLO
Tissue	L30	0.225 $\pm$.040	0.221 $\pm$.038	0.275 $\pm$.016	0.275 $\pm$.026	<u>0.270</u> $\pm$.024	0.275 $\pm$.024
	L50	0.303 $\pm$.052	0.300 $\pm$.051	<u>0.352</u> $\pm$.012	0.348 $\pm$.010	<u>0.352</u> $\pm$.023	0.354 $\pm$.019
	L70	0.344 $\pm$.034	0.326 $\pm$.020	0.393 $\pm$.023	0.392 $\pm$.024	0.404 $\pm$.018	<u>0.399</u> $\pm$.020
Derm	L30	0.419 $\pm$.007	0.417 $\pm$.009	<u>0.417</u> $\pm$.007	0.412 $\pm$.012	0.419 $\pm$.007	0.419 $\pm$.007
	L50	0.557 $\pm$.021	0.557 $\pm$.024	<u>0.561</u> $\pm$.010	0.560 $\pm$.010	<u>0.561</u> $\pm$.010	0.564 $\pm$.008
	L70	0.633 $\pm$.020	0.633 $\pm$.020	<u>0.645</u> $\pm$.018	0.641 $\pm$.015	<u>0.645</u> $\pm$.016	0.648 $\pm$.023
OctMNIST	L30	0.420 $\pm$.015	0.418 $\pm$.015	<u>0.391</u> $\pm$.023	0.326 $\pm$.034	0.388 $\pm$.022	0.407 $\pm$.019
	L50	0.564 $\pm$.031	0.565 $\pm$.030	<u>0.580</u> $\pm$.017	0.580 $\pm$.013	<u>0.581</u> $\pm$.011	0.583 $\pm$.024
	L70	0.651 $\pm$.036	0.636 $\pm$.050	<u>0.657</u> $\pm$.023	<u>0.657</u> $\pm$.022	0.658 $\pm$.021	0.650 $\pm$.026

Table 2: **The OctMNIST tight-cap win is backbone-general, grows with backbone capacity, and carries to overall accuracy.** Paired gap of TraLO over the best trained dual (per-seed best of Fioretto-LDF/Hounie-RCL) at the tight caps, mean over seeds 1–4 ($\pm$ across-seed std). Constrained-class F1 leads at every cell on all 4/4 seeds; macro-F1 improves at five of the six cells (the exception a wash at MobileNetV3 $\times$ L30), again peaking at the ViT-B/16.

Backbone	Cap	Δ cc-F1	Δ macro-F1
MobileNetV3	L30	+0.016 $\pm$.014	+0.001 $\pm$.001
	L40	+0.022 $\pm$.029	+0.008 $\pm$.013
RegNetY-400MF	L30	+0.035 $\pm$.006	+0.013 $\pm$.005
	L40	+0.021 $\pm$.003	+0.007 $\pm$.002
ViT-B/16	L30	+0.081 $\pm$.045	+0.028 $\pm$.014
	L40	+0.051 $\pm$.019	+0.016 $\pm$.008

5.2 Wins, ties, and losses across the full grid

Beyond the symmetric grid, we score the full corpus (the *paper_final* grid on the three backbones plus dedicated MobileNetV3 asymmetric sweeps), partitioned into five constraint-geometry regimes, with every regime reported so nothing is cherry-picked; the full per-cell breakdown is in Supp. Sec. D. A regime counts as a *win* only when the mean paired cc-F1 gap and at least half its cells clear +0.005, and a *loss* under the mirror-image rule; no regime mean falls below -0.005 . The pattern is consistent: **TraLO wins where the count is hard to hit and ties where it is slack**. Only two regimes clear the win bar: OctMNIST tight-symmetric caps (the backbone-general win of Sec. 5.1, largest on ViT-B/16) and the global-tighter asymmetric ($G < L$) caps of DermMNIST (+0.8pp on MobileNetV3, reproduced on RegNet and tied on ViT; *preliminary*, as only MobileNetV3 has the full asymmetric grid). Every other regime, including all symmetric caps, is a statistical tie (symmetric tissue+derm mean cc-F1 gap -0.0013 , seed winrate 29%, $n=216$), and none is a loss. Against the post-hoc clippers macro-F1 is higher for TraLO on every regime (Sec. 5.3).

5.3 Quality vs. post-hoc clipping

The other baseline family is the field default: train an unconstrained model and clip the highest-scoring predictions down to the cap. Both TraLO and clipping end up meeting the count; the question is what quality that costs. Because clipping meets the cap only by overwriting predictions (often dozens per run), it pays an accuracy tax that TraLO, landing on the count natively, avoids. Across the full 1944-run grid (run-level, paired by seed, against the best post-hoc clipper) this is a macro-F1 gain for TraLO of +0.053 (Tissue), +0.016 (Derm) and +0.021 (OctMNIST), positive in every regime and in 77 of the 81 symmetric cells (Supp. Sec. D; the exceptions are the very tightest L10/L20 caps), with constrained-class F1 moving the same way in aggregate. The margin over clipping is a property of training under the constraint rather than of TraLO alone. The dual baselines post nearly identical macro-F1 margins (the lone exception a tie at OctMNIST $\times$ ViT), so we claim it for the constraint-trained family; what is TraLO’s own is reaching that quality with the fastest and most reliable native convergence (Sec. 5.4) and the fewest prediction edits at binding caps (Supp. Sec. C). The tax is steepest where the cap binds: at the tight L30 cap, clipping costs up to 0.061 macro-F1 (Tissue; Supp. Table S2). We report how TraLO avoids the tax (near-perfect native satisfaction and an order of magnitude fewer prediction edits) as a deployment property in Supp. Sec. C: what separates the methods is the quality each preserves while meeting the count.

5.4 Convergence

These advantages all rest on one behavior, that TraLO reaches the count during training. Figure 4 tracks how far each method’s prediction count sits above the cap over the constraint phase, on the cell where the three methods separate most cleanly: DermMNIST on RegNet, whose cross-entropy saturates slowly enough that the count and the task objective openly compete. TraLO’s ratcheted multiplier pushes from the first constraint epoch and reaches the cap near epoch 12, while the zero-start dual baselines lag (Fioretto-LDF

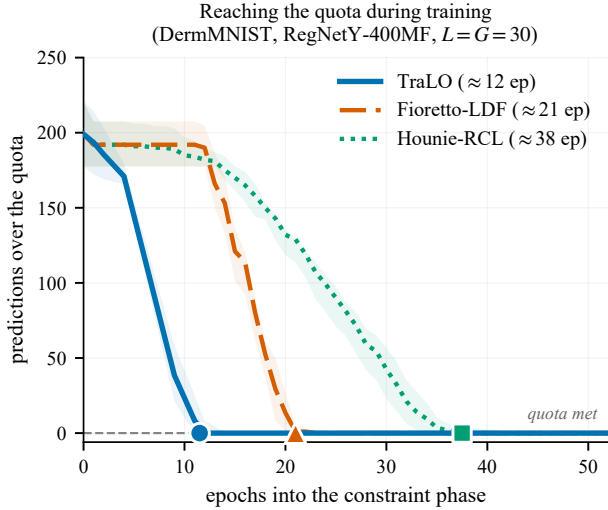


Figure 4: **TraLO reaches the count fast and holds it; the dual baselines wind up slowly.** Best-so-far constraint excess (lowest predictions over the cap so far) over the constraint phase; DermMNIST (RegNetY-400MF), tight cap $L=G=30$. Median over $n=4$ seeds; band: interquartile range; a marker sits at each method’s median epoch-to-feasibility. All three start about 200 over the cap; TraLO’s ratchet acts from the first epoch and reaches the cap first, the zero-start dual baselines later (see markers). Grid-wide numbers appear in the text.

near epoch 21, Hounie-RCL near epoch 38). The gap is not specific to this cell. Over the full grid, native feasibility is 100% for TraLO and Hounie-RCL but only 87% for Fioretto-LDF, whose unbounded multiplier stalls short of the cap on the rest. Across DermMNIST and TissueMNIST on the two convolutional backbones and the 17 symmetric-cap cells we log where the cap binds, median epochs-to-feasibility is 17/20/44 for TraLO/Fioretto-LDF/Hounie-RCL, with Hounie-RCL slowest in every cell and TraLO first or tied in 13 of the 17; the per-cell census accompanies the released corpus. TraLO alone meets the count both fast and on every run, which is what lets it avoid the post-hoc editing step that costs clipping its quality.

6 Conclusion

Count caps are a real, common deployment constraint, and the two standard answers each cost something: post-hoc clipping meets the cap but pays an accuracy tax for editing predictions, while dual-ascent training chases the cap with an unbounded multiplier and can stall short of it. TraLO addresses this with a bounded transductive penalty, a multiplier frozen at satisfaction, an optimizer reset, and an undershoot hinge. It matches the strong constraint-trained baselines across the grid and beats them on the constrained class where the cap binds hard: robustly on OctMNIST tight caps, where the lead grows with backbone capacity and reaches +0.081 on the ViT, and preliminarily at the global-tighter asymmetric caps

of DermMNIST. The ablation traces this to the optimizer reset and hinge ($n=24$ paired, $p \leq 3e-6$; Supp. Sec. B). Against post-hoc clipping it improves macro-F1 on all three datasets by landing on the count natively.

Limitations. The main limitation is scope: the cc-F1 advantage over trained baselines is regime- and dataset-specific (a tie elsewhere), and the global-tighter asymmetric win on DermMNIST is preliminary (it reproduces on RegNet but ties on the ViT, and only MobileNetV3 has the full asymmetric grid), so the one backbone-general win is OctMNIST tight caps. All three datasets are MedMNIST-derived (28×28 upsampled to 224); the count constraint itself is resolution-agnostic (it acts on prediction counts, not pixels). TraLO is also transductive: it uses the known test inputs during training, which fits deployments where the batch to be scored is fixed in advance (a triage queue, an audit pool) but not streaming or shifting-test settings, where the caps would be re-solved as the pool changes. Our analysis is also self-contained: it characterizes TraLO’s own optimization but does not *theoretically* rank it against augmented-Lagrangian, proximal, PID, or mirror-descent solvers, which would need a dedicated empirical comparison.

Broader applicability. Nothing in the formulation is specific to medical imaging: TraLO enforces a global cap on the number of inputs a classifier assigns to a class, plus per-group sub-caps, the structure of any capacity-limited decision system (a dispatch queue with a fixed budget; slots allocated across applicant groups). These are structural analogies, not validated applications; what carries over is meeting the quota during training so the model respects it natively. Validating TraLO outside medical imaging, at full resolution and wider geometries, is the natural next step.

References

- Agarwal, A.; Beygelzimer, A.; Dudík, M.; Langford, J.; and Walach, H. 2018. A Reductions Approach to Fair Classification. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, 60–69. PMLR.
- Bertsekas, D. P. 1982. *Constrained Optimization and Lagrange Multiplier Methods*. Academic Press.
- Bottou, L.; Curtis, F. E.; and Nocedal, J. 2018. Optimization Methods for Large-Scale Machine Learning. *SIAM Review*, 60(2): 223–311.
- Brosowsky, M.; Keck, F.; Düinkel, O.; and Zöllner, M. 2021. Sample-Specific Output Constraints for Neural Networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, 6812–6821.
- Cao, K.; Wei, C.; Gaidon, A.; Arechiga, N.; and Ma, T. 2019. Learning Imbalanced Datasets with Label-Distribution-Aware Margin Loss. In *Advances in Neural Information Processing Systems*, volume 32.
- Chamon, L. F. O.; Paternain, S.; Calvo-Fullana, M.; and Ribeiro, A. 2023. Constrained Learning with Non-Convex Losses. *IEEE Transactions on Information Theory*, 69(3): 1739–1760.
- Chamon, L. F. O.; and Ribeiro, A. 2020. Probably Approximately Correct Constrained Learning. In *Advances in Neural Information Processing Systems*, volume 33.

- Cortes, C.; DeSalvo, G.; and Mohri, M. 2016. Learning with Rejection. In *International Conference on Algorithmic Learning Theory (ALT)*, volume 9925 of *Lecture Notes in Computer Science*, 67–82. Springer.
- Cotter, A.; Jiang, H.; Gupta, M.; Wang, S.; Narayan, T.; You, S.; and Sridharan, K. 2019. Optimization with Non-Differentiable Constraints with Applications to Fairness, Recall, Churn, and Other Goals. *Journal of Machine Learning Research*, 20(172): 1–59.
- Cui, Y.; Jia, M.; Lin, T.-Y.; Song, Y.; and Belongie, S. 2019. Class-Balanced Loss Based on Effective Number of Samples. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 9268–9277.
- Dosovitskiy, A.; Beyer, L.; Kolesnikov, A.; Weissenborn, D.; Zhai, X.; Unterthiner, T.; Dehghani, M.; Minderer, M.; Heigold, G.; Gelly, S.; Uszkoreit, J.; and Houshy, N. 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations*.
- Eban, E.; Schain, M.; Mackey, A.; Gordon, A.; Rifkin, R.; and Elidan, G. 2017. Scalable Learning of Non-Decomposable Objectives. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, 832–840. PMLR.
- Elkan, C. 2001. The Foundations of Cost-Sensitive Learning. In *Proceedings of the Seventeenth International Joint Conference on Artificial Intelligence*, 973–978.
- Fioretto, F.; Van Hentenryck, P.; Mak, T. W. K.; Tran, C.; Baldo, F.; and Lombardi, M. 2020. Lagrangian Duality for Constrained Deep Learning. In *Machine Learning and Knowledge Discovery in Databases. Applied Data Science and Demo Track — European Conference, ECML PKDD 2020*, volume 12461 of *Lecture Notes in Computer Science*, 118–135. Springer.
- Gallego-Posada, J.; Ramirez, J.; Erraqabi, A.; Bengio, Y.; and Lacoste-Julien, S. 2022. Controlled Sparsity via Constrained Optimization or: How I Learned to Stop Tuning Penalties and Love Constraints. In *Advances in Neural Information Processing Systems*, volume 35.
- Ganchev, K.; Graça, J.; Gillenwater, J.; and Taskar, B. 2010. Posterior Regularization for Structured Latent Variable Models. *Journal of Machine Learning Research*, 11: 2001–2049.
- Ghadimi, S.; and Lan, G. 2013. Stochastic First- and Zeroth-Order Methods for Nonconvex Stochastic Programming. *SIAM Journal on Optimization*, 23(4): 2341–2368.
- Goh, G.; Cotter, A.; Gupta, M.; and Friedlander, M. P. 2016. Satisfying Real-World Goals with Dataset Constraints. In *Advances in Neural Information Processing Systems*, volume 29.
- Goyal, K.; Dumančić, S.; and Blockeel, H. 2024. DeepSaDe: Learning Neural Networks That Guarantee Domain Constraint Satisfaction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, 12199–12207.
- Hardt, M.; Price, E.; and Srebro, N. 2016. Equality of Opportunity in Supervised Learning. In *Advances in Neural Information Processing Systems*, volume 29.
- He, K.; Zhang, X.; Ren, S.; and Sun, J. 2016. Deep Residual Learning for Image Recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*, 770–778.
- Hounie, I.; Ribeiro, A.; and Chamon, L. F. O. 2023. Resilient Constrained Learning. In *Advances in Neural Information Processing Systems*, volume 36.
- Howard, A.; Sandler, M.; Chu, G.; Chen, L.-C.; Chen, B.; Tan, M.; Wang, W.; Zhu, Y.; Pang, R.; Vasudevan, V.; et al. 2019. Searching for MobileNetV3. In *IEEE/CVF International Conference on Computer Vision*, 1314–1324.
- Joachims, T. 1999. Transductive Inference for Text Classification Using Support Vector Machines. In *International Conference on Machine Learning*, 200–209.
- Lin, T.-Y.; Goyal, P.; Girshick, R.; He, K.; and Dollár, P. 2017. Focal Loss for Dense Object Detection. In *IEEE International Conference on Computer Vision*, 2999–3007.
- Mann, G. S.; and McCallum, A. 2007. Simple, Robust, Scalable Semi-supervised Learning via Expectation Regularization. In *Proceedings of the 24th International Conference on Machine Learning*, 593–600. ACM.
- Márquez-Neila, P.; Salzmann, M.; and Fua, P. 2017. Imposing Hard Constraints on Deep Networks: Promises and Limitations. *arXiv preprint arXiv:1706.02025*.
- Menon, A. K.; Jayasumana, S.; Rawat, A. S.; Jain, H.; Veit, A.; and Kumar, S. 2021. Long-Tail Learning via Logit Adjustment. In *International Conference on Learning Representations*.
- Nandwani, Y.; Pathak, A.; Mausam; and Singla, P. 2019. A Primal-Dual Formulation for Deep Learning with Constraints. In *Advances in Neural Information Processing Systems*, volume 32.
- Narasimhan, H.; Cotter, A.; and Gupta, M. 2019. Optimizing Generalized Rate Metrics with Three Players. In *Advances in Neural Information Processing Systems*, volume 32.
- Pathak, D.; Krähenbühl, P.; and Darrell, T. 2015. Constrained Convolutional Neural Networks for Weakly Supervised Segmentation. In *IEEE International Conference on Computer Vision*, 1796–1804.
- Radosavovic, I.; Kosaraju, R. P.; Girshick, R.; He, K.; and Dollár, P. 2020. Designing Network Design Spaces. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 10425–10433.
- Sangalli, S.; Erdil, E.; Hoetker, A.; Donati, O.; and Konukoglu, E. 2021. Constrained Optimization to Train Neural Networks on Critical and Under-Represented Classes. In *Advances in Neural Information Processing Systems*, volume 34, 25400–25411.
- Shifman, D. A.; Cohen, I.; Huang, K.; Xian, X.; and Singer, G. 2023. An Adaptive Machine Learning Algorithm for the Resource-Constrained Classification Problem. *Engineering Applications of Artificial Intelligence*, 119: 105741.
- Shifman, D. A.; Margolin, I.; Halfi, C.; and Singer, G. 2025. Classification Tasks with Local and Global Resource Allocation Constraints. *IFAC-PapersOnLine*, 59(1): 61–66.
- Stooke, A.; Achiam, J.; and Abbeel, P. 2020. Responsive Safety in Reinforcement Learning by PID Lagrangian Methods. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, 9133–9143. PMLR.
- Tschandl, P.; Rosendahl, C.; and Kittler, H. 2018. The HAM10000 Dataset, a Large Collection of Multi-Source Dermatoscopic Images of Common Pigmented Skin Lesions. *Scientific Data*, 5: 180161.
- Vanderschueren, T.; Baesens, B.; Verdonck, T.; and Verbeke, W. 2024. A New Perspective on Classification: Optimally Allocating Limited Resources to Uncertain Tasks. *Decision Support Systems*, 179: 114151.
- Vapnik, V. N. 1998. *Statistical Learning Theory*. Wiley.
- Yang, J.; Shi, R.; Wei, D.; Liu, Z.; Zhao, L.; Ke, B.; Pfister, H.; and Ni, B. 2023. MedMNIST v2 – A Large-Scale Lightweight Benchmark for 2D and 3D Biomedical Image Classification. *Scientific Data*, 10(1): 41.
- Zafar, M. B.; Valera, I.; Gomez Rodriguez, M.; and Gummadi, K. P. 2017. Fairness Constraints: Mechanisms for Fair Classification. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, 962–970. PMLR.